

Artificial Intelligence (AI2002)

Date: March 1st 2024

Course Instructors:

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Sessional-I Exam

Total Time: 1 Hour

Total Marks: 12.5

Total Questions: 3

Semester: Spring 2024

Campus: Karachi

Department: School of
Computing

Student Name

Roll No

Section

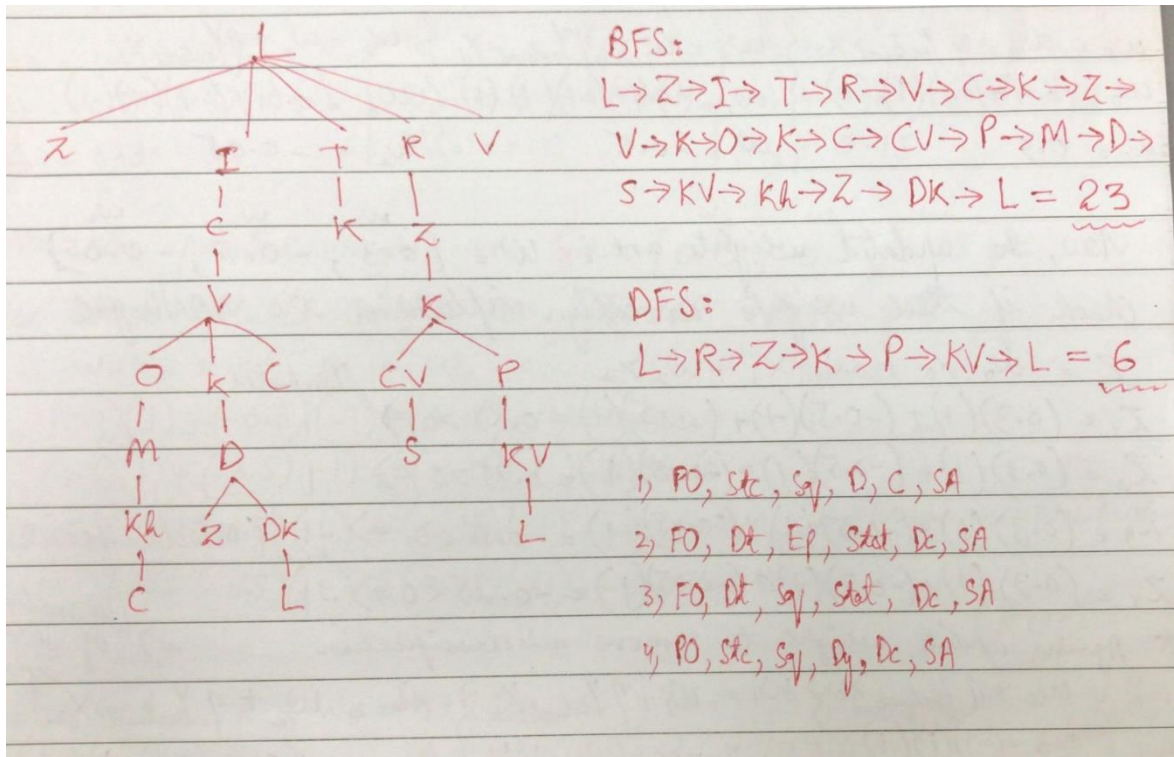
Student Signature

CLO 2: To identify and relate methods of blind as well as informed search.

- Q1 a. During the semester break, Ali plans to embark on a tour of Ukraine with his friends. His objective is to journey from the starting point in Lviv to the destination in Luhansk using the provided map. Luckily, Ali has a friend like you who is proficient in both Breadth-First Search (BFS) and Depth-First Search (DFS) algorithms. Your task is to demonstrate both algorithms by drawing the search tree and indicating the pathway (including path cost) for each strategy. Subsequently, you will analyze and determine which strategy is more effective in solving Ali's travel dilemma. **[Time: 15 mins][1.5+1.5 Marks]**



Solution:



Q1 b. When does bidirectional search perform poorly, and why? Provide an example. [Time: 2 mins][0.5 Marks]

Solution: When the graph is of a very large size with high branching factor for example a social media network, or the goal node lies in the middle.

Q1 c. When will depth-limited search perform worse as compared to the DFS? [Time: 2 mins][0.5 Marks]

Solution: Depth-limited search (DLS) performs worse than depth-first search (DFS) when the depth limit chosen for DLS is too shallow relative to the depth of the solution node in the search tree or graph.

National University of Computer and Emerging Sciences

CLO 1: To recognize the notions of rational behavior and intelligent agents.

- Q2. State the PEAS and environment types of the following: [Time: 15 mins][1+1+1+1 Marks]
1. Autonomous lawn mowing robot
 2. Home security camera
 3. Language translation app
 4. Online shopping bot

Solution: Solution can be relative according to the assumptions but it should be conceptually valid.

CLO 2: To identify and relate methods of blind as well as informed search.

- Q3 a. What do you mean by admissible and inadmissible heuristic, which one is desirable and why? [Time: 2 mins][1 Marks]

Solution:

Admissible Heuristic: A heuristic is admissible if it never overestimates the cost to reach the goal state from any given node in the search space. In other words, an admissible heuristic provides a lower bound on the actual cost to reach the goal. If $h(n)$ is an admissible heuristic for a node n , it means that $h(n) \leq h^*(n)$, where $h^*(n)$ is the true cost from node n to the goal.

Inadmissible Heuristic: Conversely, a heuristic is inadmissible if it can overestimate the cost to reach the goal state from any given node in the search space. Inadmissible heuristics can lead to suboptimal solutions in certain scenarios because they may mislead the search algorithm into exploring less promising paths.

Admissible Heuristic is desirable to ensure optimality.

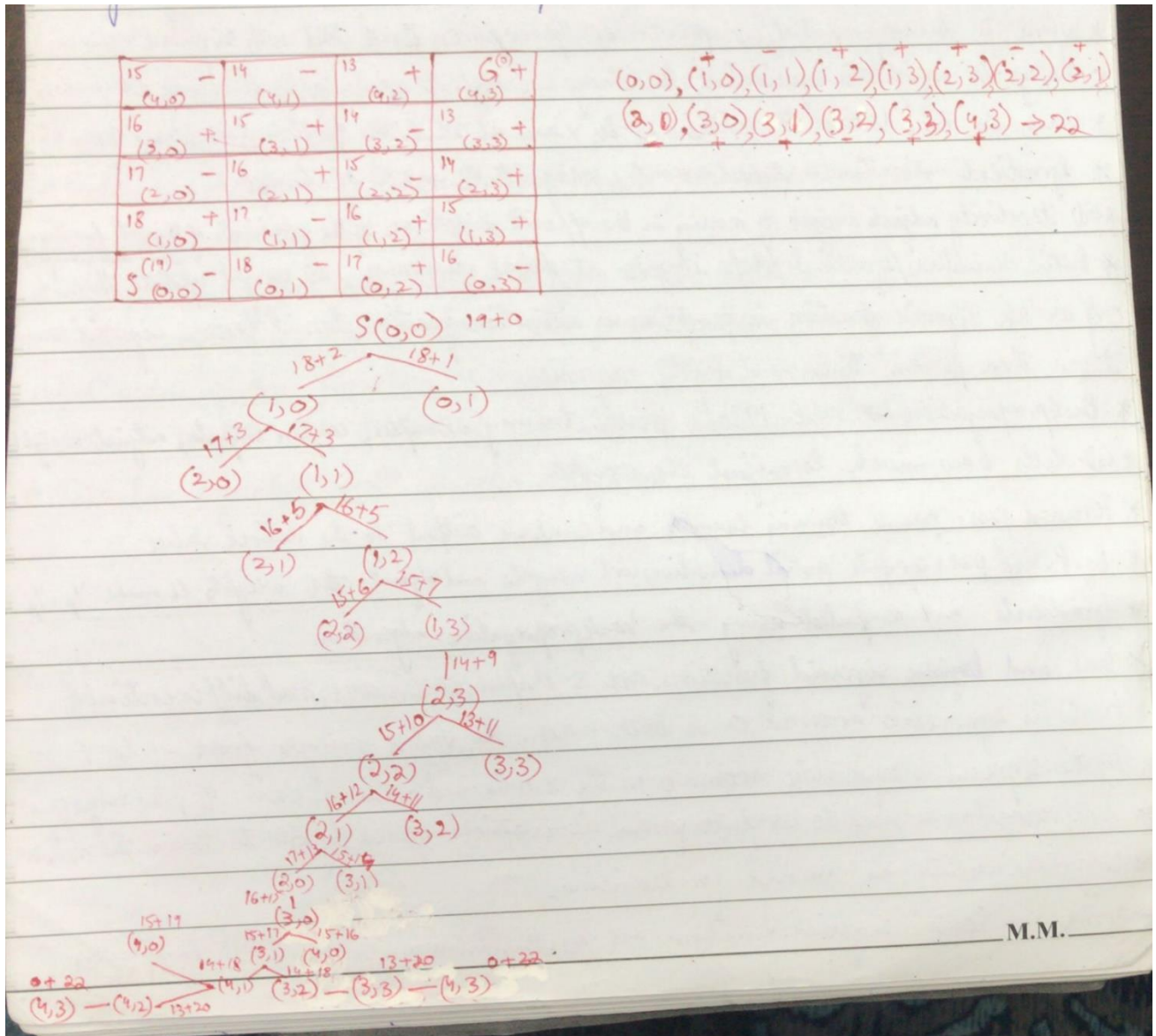
- Q3 b. A robot is deployed to navigate within a classroom environment. The robot is positioned at location S and tasked with reaching the goal state G, as depicted in the 5x4 grid below. Optimal path selection yields greater rewards. Movement is permitted in one-cell increments in the directions of Left, Right, and Up. The robot initiates at the starting state S at cell index (0,0) and must navigate toward the goal at index (4,3), with each traversed square yielding a reward of 2. However, some squares (-) offer a traversal reward of only 1.

	(4,0)	(4,1)	(4,2)	(4,3)	
(0,0)	15 -	14 -	13 +	G⁽⁰⁾ +	+ Reward = 2 - Reward = 1
(0,1)	16 +	15 +	14 -	13 +	
(0,2)	17 -	16 +	15 -	14 +	
(0,3)	18 +	17 -	16 +	15 +	
	S⁽¹⁹⁾ -	18 -	17 +	16 -	

Perform the following:

1. Apply A* search by drawing a search tree until an optimal path to the goal state is reached that maximizes the reward. The heuristic value of each cell from the goal cell is given in the figure (see box value of each cell). [Time: 15 mins][1.5 Marks]
2. Give the order in which nodes or cells will be explored by the A* algorithm. [Time: 7 mins][1.5 Marks]
3. Explain if the provided heuristic is admissible or not. [Time: 2 mins][0.5 Marks]

Solution:



Handwritten notes on a 5x4 grid showing a search process. The grid is labeled with coordinates (x,y) and values. The search starts at (0,0) and moves towards (4,3). The final path is highlighted with red arrows and numbers 12 and 13.

15 (4,0) -	14 (4,1) -	13 (4,2) +	G (4,3) +
16 (3,0) +	15 (3,1) +	14 (3,2) -	13 (3,3) +
17 (2,0) -	16 (2,1) +	15 (2,2) -	14 (2,3) +
18 (1,0) +	17 (1,1) -	16 (1,2) +	15 (1,3) +
S (0,0) -	18 (0,1) -	17 (0,2) +	16 (0,3) -

Search path (highlighted with red arrows):

- (0,0) → (1,0) → (1,1) → (1,2) → (1,3) → (2,3) → (3,3) → (4,3)

Final path (highlighted with red arrows and numbers 12 and 13):

- (0,0) → (1,0) → (1,1) → (1,2) → (1,3) → (2,3) → (3,3) → (4,3)

Handwritten notes on the right side of the page:

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[Time: 2 min
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15 +
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(0,3)

+ Reward
- Reward

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